Customer Segmentation Analysis using K-Means Clustering

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Certificate of Authenticity

This is to certify that the Mini Project 2nd entitled

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submitted by

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in partial fulfilment of the requirements for the degree of **Master of Science in Applied Mathematics** (Second Semester), Department of Mathematics, Indian Institute of Engineering Science and Technology, Shibpur, is a bonafide record of work carried out by him under my supervision and guidance.

To the best of my knowledge, the work presented in this report is original and has not been submitted elsewhere for the award of any degree, diploma, or academic qualification.

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This project was a genuinely interesting experience. Working with a real dataset and seeing how a mathematical model could actually reveal hidden structure felt quite rewarding.

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Abstract

This project applies the K-Means clustering algorithm to the Mall Customer Segmentation Dataset to partition 200 customers into groups based on their annual income and spending behaviour, with the goal of understanding both the mathematical structure of the algorithm and the practical meaning of the clusters it produces.

K-Means minimises the Within-Cluster Sum of Squares (WCSS) by alternating between two steps: assigning each data point to its nearest centroid via Euclidean distance, and recomputing centroids as the arithmetic mean of the assigned points. Both steps are non-increasing in WCSS, guaranteeing convergence. Two features — Annual Income (k$) and Spending Score (1–100) — were selected, standardised, and used as input. The optimal cluster count was chosen via the Elbow Method, which revealed a sharp kink at $k = 5$.

The five segments correspond to recognisable spending profiles: premium customers (high income, high spending), cautious high earners (high income, low spending), impulsive low-income shoppers, budget visitors (low income, low spending), and a central average segment. These groupings carry direct implications for targeted marketing strategies. The analysis is implemented entirely in R, with emphasis on conceptual understanding over coding complexity.

Keywords: K-Means Clustering, Customer Segmentation, WCSS, Elbow Method, Unsupervised Learning, Mall Customer Dataset.

Contents

1. Introduction

Businesses collect a large amount of information about their customers — purchase history, income levels, frequency of visits, and so on — yet often struggle to use this data in a structured way. One fundamental step in making sense of customer data is segmentation: dividing customers into groups that share similar characteristics. Rather than treating every customer identically, segmentation allows a company to focus its strategies more precisely, whether for product recommendations, pricing, or targeted promotions.

Customer segmentation is not a new concept. Before modern computing, marketers relied on surveys and demographic categories to create broad customer profiles. What has changed is the ability to process large datasets and identify patterns that would otherwise remain invisible to manual inspection. Machine learning, particularly unsupervised methods, offers a principled way to perform this grouping without needing pre-labelled data.

Among the various clustering techniques available, K-Means stands out for its simplicity and computational efficiency. It operates on a clear geometric idea: customers who are close to each other in some feature space should belong to the same group. The algorithm defines closeness through Euclidean distance and iteratively refines cluster assignments until the total within-group variation is minimised.

The motivation behind this project is to understand this process in a concrete, mathematically grounded way. Rather than applying the method as a black box, the goal is to work through the mathematics, understand what the algorithm is actually optimising, and then interpret the results in a way that makes practical sense. The Mall Customer Segmentation Dataset provides a small, clean dataset that is well-suited for this kind of exploratory analysis.

2. Objective of the Study

The objectives of this project can be stated as follows:

1. To understand the theoretical basis of the K-Means clustering algorithm from a mathematical perspective.

2. To identify a suitable number of clusters using the Elbow Method based on WCSS analysis.

3. To apply K-Means clustering on the Mall Customer Segmentation Dataset and group customers according to their annual income and spending score.

4. To interpret the resulting clusters in a practically meaningful way.
5. To understand the limitations of K-Means and recognise situations where other approaches might be preferable.

3. Basic Concepts of Machine Learning

3.1. What is Machine Learning?

Machine learning is a branch of artificial intelligence that deals with building algorithms capable of learning from data. Rather than encoding rules explicitly, a machine learning model identifies patterns in a training set and uses those patterns to make predictions or decisions on new data. Broadly, machine learning methods are divided into supervised learning, unsupervised learning, and reinforcement learning.

3.2. Supervised Learning

In supervised learning, the training data comes with labelled outputs. The model learns a mapping from inputs $\mathbf{x} \in \mathbb{R}^p$ to outputs y, where y is a class label (in classification) or a real value (in regression). Formally, we seek a function $f : \mathbb{R}^p \to \mathcal{Y}$ that minimises a specified loss between predicted and true outputs. Linear regression and decision trees are simple examples of this category.

3.3. Unsupervised Learning

Unsupervised learning, by contrast, works with data that carries no labels. The task is to discover some inherent structure in the data itself. This is more open-ended than supervised learning — the algorithm is not told what to look for. Common tasks include clustering (grouping similar observations), dimensionality reduction (finding compact representations), and density estimation.

Mathematically, let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \in \mathbb{R}^p$ be n unlabelled data points. The goal is to assign each $\mathbf{x}_i$ to one of k groups such that some measure of intra-group similarity is maximised (or intra-group dissimilarity is minimised).

3.4. Clustering

Clustering is the most widely used task in unsupervised learning. A clustering algorithm partitions the dataset into k subsets, or clusters, $C_1, C_2, \dots, C_k$, such that:

$$C_i \cap C_j = \emptyset \quad \forall\, i \neq j, \qquad \bigcup_{i=1}^{k} C_i = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}.$$

Points within a cluster are expected to be more similar to each other than to points in other clusters, where similarity is typically measured through a distance function.

Clustering has applications in document classification, image segmentation, anomaly detection, and, as in this project, customer segmentation.

4. K-Means Clustering

4.1. Basic Idea

K-Means is a partition-based clustering algorithm. Given a dataset of n points in $\mathbb{R}^p$ and a pre-specified number of clusters k, the algorithm attempts to place each data point into one of k groups in a way that minimises the total within-cluster variance. The name comes from the fact that each cluster is represented by the mean (centroid) of its members.

4.2. Centroid

The centroid of a cluster C_j is simply the arithmetic mean of all points assigned to that cluster:

$$\boldsymbol{\mu}_j = \frac{1}{|C_j|} \sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i,$$

where $|C_j|$ denotes the number of points in C_j. The centroid acts as a representative location for the cluster — a point that minimises the sum of squared distances to all members of that cluster.

4.3. The Iterative Process

The algorithm proceeds in two alternating steps:

Assignment Step. Each point $\mathbf{x}_i$ is assigned to the cluster whose centroid is nearest in terms of Euclidean distance:

$$z_i = \arg \min_{j \in \{1,\dots,k\}} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2.$$

Update Step. Once all points have been reassigned, each centroid is recomputed as the mean of the newly formed cluster.

These two steps alternate until the cluster assignments no longer change, i.e., until convergence is reached. It is guaranteed to converge in a finite number of iterations (since there are finitely many partitions), though the solution may be a local rather than global minimum of the objective function.

4.4. Algorithm Summary

The complete procedure can be stated as follows:

1. Choose k, the number of clusters.
2. Initialise k centroids $\boldsymbol{\mu}_1^{(0)}, \ldots, \boldsymbol{\mu}_k^{(0)}$, often by selecting k points randomly from the dataset (K-Means++ improves this step).
3. **Repeat until convergence:**
   a. Assign each $\mathbf{x}_i$ to the nearest centroid: $z_i \leftarrow \arg\min_j \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$.
   b. Update centroids: $\boldsymbol{\mu}_j \leftarrow \frac{1}{|C_j|} \sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i$.
4. Return the final cluster assignments $\{z_i\}$ and centroids $\{\boldsymbol{\mu}_j\}$.

5. Mathematical Background

5.1. Euclidean Distance

The foundation of K-Means is the notion of distance between two points. For two vectors $\mathbf{u} = (u_1, u_2, \ldots, u_p)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_p)$ in $\mathbb{R}^p$, the Euclidean distance is:

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\sum_{l=1}^{p} (u_l - v_l)^2}. \tag{1}$$

In the context of clustering, $\mathbf{u}$ is a data point and $\mathbf{v}$ is the centroid of a candidate cluster. Squaring this distance (i.e., using d^2 instead of d) is computationally convenient and does not change which centroid is nearest, since the square root is a monotone function.

Intuition. Equation (1) is essentially the generalisation of the Pythagorean theorem to p dimensions. In the two-dimensional case ($p = 2$), it reduces to the familiar formula $\sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}$. For our dataset, $p = 2$ (income and spending score), so the geometry remains directly visualisable.

5.2. Centroid as the Minimiser of Squared Distance

It is worth noting that the centroid is not just defined as the mean — it is, in fact, the point that minimises the sum of squared distances to all members of the cluster. For a cluster C with members $\mathbf{x}_1, \ldots, \mathbf{x}_m$, we want to find $\boldsymbol{\mu}$ that minimises:

$$f(\boldsymbol{\mu}) = \sum_{i=1}^{m} \|\mathbf{x}_i - \boldsymbol{\mu}\|^2.$$

Taking the derivative and setting it to zero:

$$\frac{\partial f}{\partial \boldsymbol{\mu}} = -2\sum_{i=1}^{m}(\mathbf{x}_i - \boldsymbol{\mu}) = \mathbf{0} \implies \boldsymbol{\mu} = \frac{1}{m}\sum_{i=1}^{m}\mathbf{x}_i.$$

This confirms that the arithmetic mean is the natural representative of a cluster under a squared-Euclidean criterion.

5.3. The Objective Function

The global quantity that K-Means seeks to minimise is the **Within-Cluster Sum of Squares (WCSS)**, also called the inertia:

$$J = \sum_{j=1}^{k}\sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2. \tag{2}$$

Here, the outer sum runs over the k clusters, and the inner sum runs over all points assigned to cluster C_j. Each term $\|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$ measures how far point $\mathbf{x}_i$ is from the centre of its cluster.

Intuition. A small value of J means that points within each cluster are tightly packed around their centroid, which corresponds to a compact, well-separated clustering. As k increases, J decreases (in the extreme case $k = n$, each point forms its own cluster and $J = 0$), which is why choosing k based on J alone is not meaningful — we need an additional criterion.

5.4. Decomposition of Total Variance

For a single cluster C_j with centroid $\boldsymbol{\mu}_j$, the within-cluster sum of squares can be written as:

$$\mathrm{WCSS}_j = \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2 = \sum_{l=1}^{p} \mathrm{Var}_l(C_j) \cdot |C_j|, \tag{3}$$

where $\mathrm{Var}_l(C_j)$ is the variance of the l-th coordinate within cluster C_j. This shows that minimising WCSS is equivalent to minimising the total coordinate-wise variance across all clusters, which is a familiar statistical objective.

5.5. Convergence

Each iteration of the algorithm does not increase J. The assignment step can only decrease J (since each point moves to a closer centroid), and the update step can also only decrease J (since the mean minimises the sum of squared distances within a cluster). Since J is bounded below by zero and is a non-increasing sequence of non-negative integers over a finite set of partitions, the algorithm must converge. However, it may converge to a local, not global, minimum. Running the algorithm multiple times with different initialisations (controlled by `nstart` in R) is a standard practical remedy.

6. Advantages and Limitations of K-Means

6.1. Advantages

- **Simplicity.** The algorithm is easy to understand and implement. The core idea — group points by proximity to their cluster mean — requires no advanced probabilistic machinery.

- **Scalability.** K-Means has a time complexity of $O(nkpt)$, where t is the number of iterations. It handles moderate to large datasets efficiently, unlike methods such as hierarchical clustering that require $O(n^2)$ distance computations.

- **Interpretability.** Cluster centroids have a concrete meaning — they are the average customer in that group — making the results relatively easy to explain to a non-technical audience.

- **Convergence guarantee.** The objective function is non-increasing across iterations, so the algorithm always terminates in finite time.

6.2. Limitations

- **Requires k in advance.** The number of clusters must be specified before running the algorithm. Poor choice of k leads to either over-segmented or under-segmented results.

- **Sensitivity to initialisation.** Different starting centroids can lead to different final clusters. The algorithm may converge to a suboptimal local minimum.

- **Assumes spherical clusters.** K-Means uses Euclidean distance, which implicitly assumes that clusters are roughly convex and of similar size. It performs poorly on elongated, ring-shaped, or non-convex clusters.

- **Sensitivity to outliers.** Since centroids are means, a single extreme point can pull the centroid away from the rest of the cluster, distorting the grouping.

- **Scaling matters.** Features with larger numerical ranges dominate the distance computation unless the data is standardised beforehand.

7. Dataset Description

The dataset used in this project is the **Mall Customer Segmentation Dataset**, a commonly used dataset for introductory clustering tasks. It is available on Kaggle and contains records for 200 customers of a supermarket mall.

7.1. Variables

The dataset has five columns, described in Table 1.

Table 1. Description of variables in the Mall Customer Segmentation Dataset.

Col.	Variable Name	*Type*	Description
1	CustomerID	*Integer*	Unique identifier for each customer
2	Gender	*Categorical*	Male or Female
3	Age	*Integer*	Age of the customer (years)
4	Annual Income	*Continuous*	Annual income in thousands of USD
5	Spending Score	*Integer*	Score assigned by the mall (1–100)

7.2. Practical Meaning of Variables

The most analytically interesting variables are **Annual Income** and **Spending Score**. Annual income reflects the economic capacity of a customer, while the spending score is an internally assigned metric based on purchase behaviour and frequency — a higher score indicates a more active or higher-spending customer.

The combination of these two variables allows for meaningful groupings: a customer with high income and a high spending score behaves very differently from one with high income but a low score (perhaps a conservative saver), or one with low income but a high score (possibly an impulsive spender).

7.3. Dataset Structure

A brief numerical summary of the key variables is given below:

Table 2. Summary statistics for selected variables.

Variable	Min	Max	Mean	Median	Std. Dev.
Age	18	70	38.85	36.0	13.97
Annual Income	15	137	60.56	61.5	26.26
Spending Score	1	99	50.20	50.0	25.82

The nearly symmetric distribution of spending scores around a mean of 50 suggests that the mall's scoring system is reasonably calibrated. Annual income, on the other hand, shows a somewhat wider range, and one would expect income-spending combinations to vary considerably across the 200 customers.

8. Data Preprocessing

8.1. Feature Selection

For this analysis, only two variables are used: Annual Income (k$) and Spending Score. The CustomerID column is a simple identifier and carries no information. Gender and Age, while potentially useful in a richer analysis, are excluded here to keep the clustering two-dimensional and directly visualisable.

Selecting a two-dimensional feature space means that each customer is represented as a point $(x_1, x_2) \in \mathbb{R}^2$, where x_1 is the annual income and x_2 is the spending score. This allows the final clusters to be plotted directly, which aids interpretation.

8.2. Handling Missing Values

The dataset is clean and contains no missing values, so no imputation is required. In a real-world setting, this would typically be one of the first checks performed.

8.3. Feature Scaling

Annual income ranges from 15 to 137 (in k$) while spending score ranges from 1 to 99. Although the scales are not drastically different here, standardisation is still advisable to prevent either variable from inadvertently dominating the distance computations in (1).

Standardisation transforms each variable to have zero mean and unit variance:

$$x_l^* = \frac{x_l - \bar{x}_l}{\sigma_l}, \tag{4}$$

where $\bar{x}_l$ and σ_l are the sample mean and standard deviation of the l-th feature. After this transformation, both variables contribute roughly equally to the Euclidean distance. In R, this is handled by the `scale()` function, which applies (4) column-wise.

9. Elbow Method

9.1. Why Choosing k Matters

K-Means requires the number of clusters k to be fixed before the algorithm runs. Choosing k too small merges genuinely different customer groups; choosing it too large creates artificial distinctions that are harder to interpret and act on. The Elbow Method is a simple heuristic to guide this choice.

9.2. WCSS and the Elbow

Recall that the objective function J in (2) measures total within-cluster dispersion. As k increases from 1 to n:

- For small k, each cluster is large and heterogeneous, so J is large.
- As k grows, clusters become more homogeneous and J decreases.
- At $k = n$, every point is its own cluster and $J = 0$.

The rate of decrease in J is typically steep at first and then levels off. The **Elbow Method** plots J against k and looks for the value of k at which the decrease becomes noticeably less steep — visually resembling the bend of an elbow. This point represents a reasonable balance between compactness (small J) and parsimony (small k).

Formally, the point of inflection can be thought of as the k where:

$$\Delta J(k) = J(k-1) - J(k) \gg J(k) - J(k+1),$$

meaning the marginal gain in compactness from adding one more cluster drops off sharply.

9.3. Limitations of the Elbow Method

Note on Limitations

The elbow is not always clearly defined — sometimes the WCSS curve decreases smoothly without a sharp kink. In such cases, methods like the **Silhouette Score** or the **Gap Statistic** may provide clearer guidance. For the Mall Customer Dataset, the elbow at $k = 5$ is relatively distinct and consistently observed.

10. Model Implementation in R

The full R code is included in the Appendix. Here, only the key steps are presented for clarity.

10.1. Loading and Preparing the Data

```r
# Load the dataset
mall_data <- read.csv("Mall_Customers.csv")

# Select Annual Income and Spending Score
```

```r
features <- mall_data[, c("Annual.Income..k..", "Spending.Score
    ..1.100.")]

# Scale the features
features_scaled <- scale(features)
```

Listing 1. Loading and selecting features

10.2. Elbow Method

```r
wcss <- numeric(10)
for (k in 1:10) {
  km <- kmeans(features_scaled, centers = k, nstart = 25)
  wcss[k] <- km$tot.withinss
}

# Plot the elbow curve
plot(1:10, wcss, type = "b", pch = 19,
     xlab = "Number of Clusters (k)",
     ylab = "WCSS",
     main = "Elbow Method for Optimal k",
     col = "steelblue")
```

Listing 2. Elbow method: computing WCSS for k = 1 to 10

10.3. Applying K-Means

```r
set.seed(42)
km_model <- kmeans(features_scaled, centers = 5, nstart = 25)

# Cluster assignments
mall_data$Cluster <- as.factor(km_model$cluster)
```

Listing 3. K-Means with k = 5

10.4. Visualising the Clusters

```r
library(ggplot2)

ggplot(mall_data, aes(x = Annual.Income..k..,
                      y = Spending.Score..1.100.,
                      color = Cluster)) +
  geom_point(size = 3, alpha = 0.8) +
  labs(title = "Customer Segments (K-Means, k = 5)",
       x = "Annual Income (k$)",
       y = "Spending Score (1-100)") +
  theme_minimal()
```

Listing 4. Scatter plot of clusters

11. Results and Interpretation

11.1. Elbow Plot

Figure 1 (schematic) shows the WCSS plotted against k for $k = 1$ to 10. The curve decreases sharply from $k = 1$ to $k = 5$, after which the rate of decrease becomes much smaller. This suggests that adding a sixth or seventh cluster brings relatively little improvement in compactness. The elbow appears at $k = 5$, which is taken as the optimal number of clusters.

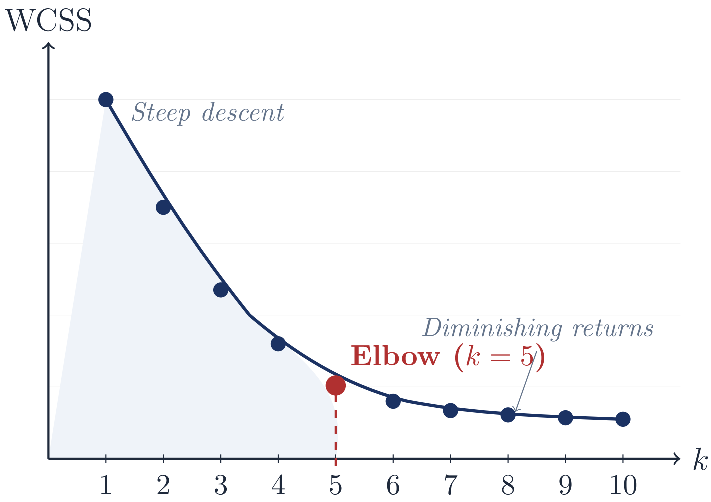

Figure 1. Schematic of the Elbow plot. The sharpest change in slope occurs at $k = 5$.

11.2. Cluster Plot

Figure 2 (schematic) illustrates the five clusters in the income–spending score plane. The five groups are visually well-separated, which reflects the relatively clean structure of this particular dataset.

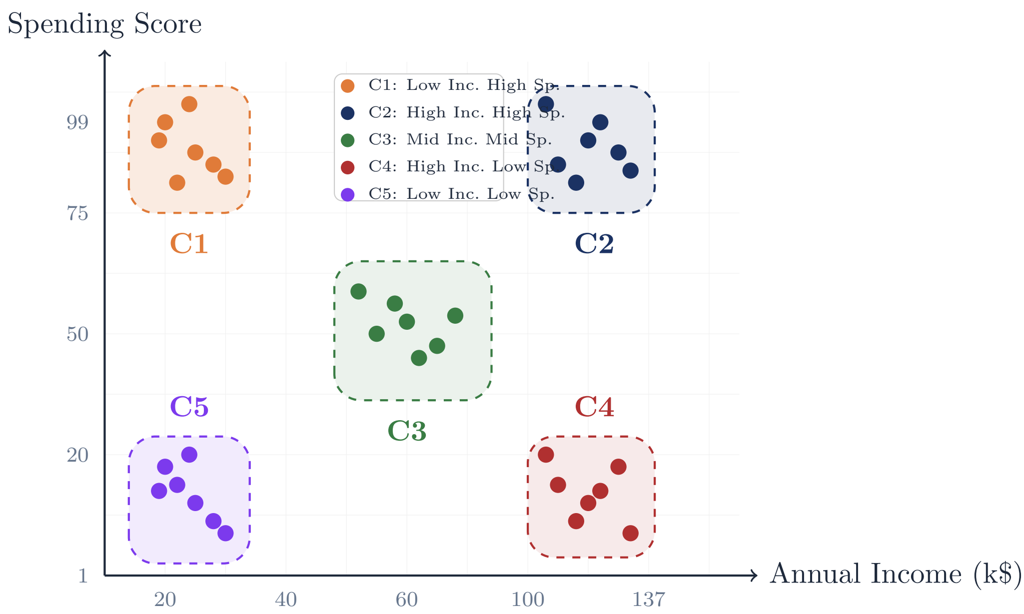

Figure 2. Schematic representation of the five customer clusters in the income–spending score plane.

11.3. Interpretation of Customer Groups

The five clusters can be described as follows, based on their approximate positions in the income–spending plane:

Table 3. Interpretation of the five customer clusters.

Cluster	Income Level	Spending Level	Profile
C1	Low	High	Active low-income shoppers
C2	High	High	Ideal / Premium customers
C3	Medium	Medium	Standard / Typical customers
C4	High	Low	Wealthy but conservative shoppers
C5	Low	Low	Budget-conscious, infrequent visitors

Cluster C2 (high income, high spending) represents the most desirable customer segment for the mall from a revenue perspective. These are customers who have the financial capacity to spend generously and actively choose to do so.

Cluster C4 (high income, low spending) is particularly interesting: despite having above-average earnings, these customers spend sparingly. This could indicate dissatisfaction with the mall's offerings, a preference for saving, or simply that the mall is not their primary shopping destination.

Cluster C1 (low income, high spending) captures a group of enthusiastic shoppers who may be spending above their financial comfort zone. For the mall management, this group requires some caution — over-relying on promotions that encourage further expenditure here may not be sustainable.

Cluster C3 is the largest and most diffuse group. These are the "average" customers — neither particularly high-value nor particularly disengaged.

Cluster C5 (low income, low spending) consists of customers who visit rarely and spend little. Targeted outreach (value deals, discount coupons) might gradually shift some members of this cluster into a more engaged segment.

12. Discussion

The results of this analysis illustrate how a straightforward clustering procedure can surface meaningful structure in customer data. The five-segment model aligns well with the kind of customer profiling that marketing teams in retail businesses routinely seek. The mathematical underpinning — minimising squared Euclidean distances to cluster centroids — provides a principled basis for what might otherwise be a subjective grouping exercise.

From a practical standpoint, the most actionable finding is the distinction between clusters C2 and C4. Both represent high-income customers, but their spending behaviour differs substantially. Understanding this gap — whether it stems from product range, pricing, parking convenience, or brand perception — is a question that K-Means cannot answer on its own, but the segmentation at least raises it sharply.

It is worth being explicit about what K-Means is and is not. It does not provide a probabilistic measure of cluster membership, meaning it gives no sense of how confidently a customer belongs to a given segment versus a neighbouring one. Soft clustering methods, such as Gaussian Mixture Models, assign each customer a probability of belonging to each cluster, which can be more informative for borderline cases.

The choice of Euclidean distance also assumes implicitly that all features are equally scaled and that clusters are convex. Both of these assumptions were partially addressed by scaling the data, but the spherical cluster assumption remains. In this particular dataset, the clusters happen to be fairly compact and separated, so K-Means works well. In datasets where clusters have irregular shapes or wildly different densities, DBSCAN or spectral clustering would be preferable.

13. Conclusion

This project applied K-Means clustering to the Mall Customer Segmentation Dataset, using Annual Income and Spending Score as the two clustering features. The Elbow Method suggested $k = 5$ as the appropriate number of clusters, and the resulting segmentation produced five interpretable customer groups that differ meaningfully in their income and spending profiles.

From a mathematical standpoint, the project demonstrated how an iterative optimisation procedure — alternating between point-to-centroid assignment and centroid recomputation — converges to a local minimum of the WCSS objective. The connection between the centroid formula and the least-squares minimiser highlights that K-Means is not an arbitrary heuristic, but a well-defined optimisation algorithm grounded in standard linear algebra.

The analysis also highlighted the limitations of the method: sensitivity to initial conditions, the assumption of spherical clusters, and the need to specify k in advance. These are not reasons to dismiss the algorithm, but important caveats to keep in mind when applying it to more complex real-world datasets.

Overall, K-Means provides a clean, interpretable starting point for customer segmentation tasks, and the Mall Customer Dataset offers a concrete setting in which its strengths and weaknesses can be observed directly.

14. Future Scope

Several natural extensions of this work present themselves:

- **Incorporating more features.** The current analysis uses only two variables. Including age, gender, and purchase frequency would give a richer representation of customer behaviour and might reveal finer-grained segments.

- **Alternative clustering methods.** Gaussian Mixture Models (GMMs) allow soft cluster assignments and can model elliptical clusters more flexibly. DBSCAN is useful for datasets where clusters are non-convex or have varying densities. Hierarchical clustering offers a dendrogram-based view that does not require k to be fixed in advance.

- **Cluster validity indices.** Beyond the Elbow Method, measures such as the Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index provide a more rigorous basis for evaluating clustering quality.

- **Larger retail datasets.** Applying this framework to transactional data (e.g., RFM analysis — Recency, Frequency, Monetary value) in a full-scale retail context would be a more realistic business application.

- **Hybrid approaches.** Dimensionality reduction techniques such as PCA could be applied before clustering to denoise high-dimensional data, potentially improving the quality of K-Means results on larger feature sets.

A. Complete R Code

The following code reproduces all results reported in this project.

```
# --------------------------------------------------------
# Customer Segmentation using K-Means Clustering
# M.Sc Applied Mathematics Mini Project
# --------------------------------------------------------

# 1. Load required libraries
library(ggplot2)

# 2. Load the dataset
mall_data <- read.csv("Mall_Customers.csv")

# 3. Inspect the data
head(mall_data)
summary(mall_data)

# 4. Select relevant features
features <- mall_data[, c("Annual.Income..k..",
                           "Spending.Score..1.100.")]

# 5. Scale the features (zero mean, unit variance)
features_scaled <- scale(features)

# 6. Elbow Method: compute WCSS for k = 1 to 10
wcss <- numeric(10)
for (k in 1:10) {
  set.seed(42)
  km <- kmeans(features_scaled, centers = k,
               nstart = 25, iter.max = 300)
  wcss[k] <- km$tot.withinss
}

# 7. Plot the Elbow curve
plot(1:10, wcss, type = "b", pch = 19, col = "steelblue",
     xlab = "Number of Clusters (k)",
     ylab = "Within-Cluster Sum of Squares (WCSS)",
     main = "Elbow Method for Optimal k")
abline(v = 5, col = "red", lty = 2)

# 8. Fit K-Means with k = 5
set.seed(42)
km_model <- kmeans(features_scaled, centers = 5,
                   nstart = 25, iter.max = 300)

# 9. Print cluster summary
```

```r
print(km_model$centers)   # Scaled centroids
cat("Cluster sizes:", km_model$size, "\n")
cat("Total WCSS:", km_model$tot.withinss, "\n")

# 10. Append cluster labels to original data
mall_data$Cluster <- as.factor(km_model$cluster)

# 11. Visualise the clusters on original scale
ggplot(mall_data,
       aes(x = Annual.Income..k..,
           y = Spending.Score..1.100.,
           color = Cluster)) +
  geom_point(size = 3.5, alpha = 0.85) +
  labs(
    title = "Customer Segments -- K-Means (k = 5)",
    x     = "Annual Income (k$)",
    y     = "Spending Score (1-100)",
    color = "Cluster"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title      = element_text(hjust = 0.5, face = "bold"),
    legend.position = "right"
  )

# 12. Cluster-wise summary statistics
aggregate(mall_data[, c("Annual.Income..k..",
                         "Spending.Score..1.100.")],
          by  = list(Cluster = mall_data$Cluster),
          FUN = mean)
```

Listing 5. Complete R code for customer segmentation analysis

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